SEUNGWON JEONG

Melbourne Connect, 700 Swanston Street, Carlton, VIC 3053, Australia
(+61) 431-217-210 ◇ seungwon.jeong@student.unimelb.edu.au
svvj.github.io ◇ github.com/svvj ◇ Google Scholar

EDUCATION

University of Melbourne *Aug 2024 - Present*
Doctor of Philosophy (PhD Candidate)
Research: Large-space modeling and rendering for free-view navigation

Korea University *Mar 2022 - Feb 2024*
Master of Science, Artificial Intelligence Applications GPA: 4.11/4.5 (95.5%)
Research Topic: Computer Graphics, Physics-Based Simulation
Thesis: A Hybrid Iterative Algorithm for Solving Constraint Optimization Problems

Korea University *Mar 2016 - Feb 2022*
Bachelor of Engineering, Computer Science and Engineering GPA: 3.65/4.5 (91.5%)
Interdisciplinary Major in Artificial Intelligence (subsidiary)

Daejeon Science High School *Mar 2013 - Feb 2016*
Daejeon, South Korea

RESEARCH INTERESTS

Computer Graphics, 3D Gaussian Splatting, Neural Rendering, Computer Vision, Free-view Navigation, XR/VR/AR, Visual Effects (VFX), and Human-Computer Interaction.

PUBLICATIONS

Flashover: Spatial Storytelling of Wildfire from Memory to Spatial Experience.
Phillip Wilkinson, Robert Ellis Walton, SeungWon Jeong, Taehyun Rhee.
ACM SIGGRAPH 2026 (Experience Hall / Spatial Storytelling). Program

Perceptual Thresholds of Sample Counts in AI-Augmented 3D Gaussian Splatting.
SeungWon Jeong, Lingxiao Li, Bin Chen, Taehyun Rhee.
IEEE VR Abstracts and Workshops (VRW), 2026. DOI

Free-View XR Tours Using 3D Gaussian Splatting with an Avatar Guide.
Mayura Manawadu, SeungWon Jeong, Lingxiao Li, Bin Chen, Taehyun Rhee.
IEEE VR Abstracts and Workshops (VRW), 2026. DOI

How Many Are Enough? Effective Sample Counts in NeRF and 3DGS.
SeungWon Jeong, Bin Chen, Taehyun Rhee.
IEEE ISMAR-Adjunct, 2025. DOI

The Impact of Occlusion and Collision on Real–Virtual Interaction in Mixed Reality 360° Images.
Andrew Chalmers, SeungWon Jeong, Xuteng Yuan, Fanglue Zhang, Taehyun Rhee.
IEEE ISMAR-Adjunct, 2025. DOI

Superpixel-guided Sampling for Compact 3D Gaussian Splatting.
Myoung Gon Kim, SeungWon Jeong, Seohyeon Park, JungHyun Han.
ACM Symposium on Virtual Reality Software and Technology (VRST), 2024. DOI

A Hybrid Iterative Algorithm for Solving Constraint Optimization Problems.
SeungWon Jeong. Master's Thesis, Korea University (Advisor: JungHyun Han), 2024.

WORK EXPERIENCE

University of Melbourne 2025 & 2026
Teaching Assistant / Tutor *Part-time*

- Graphics and Interaction (COMP30019), Semester 2 of 2025 and 2026: run tutorials and consultations on real-time 2D/3D graphics, Unity, and shader programming, and deliver assessment and feedback for project-based coursework.

Next-generation VR/AR Research Institute, Korea University Mar 2024 - Aug 2024
Research Engineer *Full-time*

- Led research improvements for the Digital Human project.
- Developed physics-based and learning-based components for virtual humans and clothing simulation.

Computer Graphics Course, Korea University Mar 2022 - Jul 2022
Teaching Assistant *Part-time*

- Designed OpenGL-based programming assignments on character animation and ray tracing.

Joanholab Co. Nov 2019 - Jun 2020
Backend Developer *Part-time*

- Led backend development, login authentication, database management and deployment.

Republic of Korea Army Mar 2018 - Nov 2019
Compulsory Military Service *Full-time*

- Served as a tactical server administrator; supported server management and communication tasks.

PROJECTS

Flashover: Spatial Storytelling of Wildfire *University of Melbourne, 2025–2026*
Immersive multi-zone VR installation recreating firefighters' Black Summer bushfire memories with point-based volumetric imagery, **in close collaboration with the Faculty of Fine Arts and Music (FFAM)**; exhibited at ACM SIGGRAPH 2026 Experience Hall. *(Unity, VR/XR)*

Digital Human Project *Korea University, 2023–2024*
Real-time virtual human and clothing from RGB footage, combining physics-based simulation with deep learning. *(Python, PyTorch, Taichi, OpenGL)*

Physics-Based Simulation Framework *Korea University, 2023*
Simulation framework with OBJ I/O, a custom renderer, and numerical solvers for constraint optimization. *(C++, OpenGL)*

Differentiable Flag *Korea University, 2021–2022*
Differentiable physics simulation that reconstructs flag motions from RGB video and 3D reconstructions. *(Python, PyTorch, PyTorch3D)*

Outsourcing Website Production *Joanholab Co., 2019–2020*
Led backend login/authentication, database management, and deployment as Web Development Team Leader. *(TypeScript, Node.js, MongoDB, Docker)*

AWARDS & SCHOLARSHIPS

Best Research Award *2023*
Professional Manpower Training for VR·AR Research Contest, Korea Electronics Association.

Melbourne Research Scholarship (Stipend and Fee Offset) *2024 - Present*
100% fee remission and living allowance (AUD$39,500/year pro rata), University of Melbourne.

Student Creative and Independent Project Grant (about USD$7,500) *2022*
ITRC Support Program, Ministry of Science and ICT, Korea (ICT Creative Consilience Program).

Dang-Rim Scholarship (about USD$2,300) *2021*

Full Scholarship Throughout Undergraduate Studies (about USD$13,000 total, National Grant) *2016–2017, 2020–2021*

TECHNICAL STRENGTHS

Programming	C++, C, Python, CUDA, C#, JavaScript, TypeScript
Graphics & ML	OpenGL, Shaders (GLSL/HLSL), PyTorch, OpenCV, Taichi
Game Engine & XR	Unity, VR/XR development, OpenXR
Web & Data	Node.js, MongoDB
Tools & Env.	Linux, Windows, Docker, Git, Visual Studio, VS Code, PyCharm, CLion
Creative	Blender, Photoshop, Premiere Pro, LaTeX

LANGUAGES

Korean	Mother tongue
English	TOEFL iBT 88